Finite Prime-Moduli Elliptic Signatures for MR/US Fusion and Bounded Intraoperative Energy Envelopes

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ABSTRACT

We describe a computational MR/US fusion demonstrator that co-registers synthetic pre-operative T2-weighted MR anatomy with a synthetic 8.5 MHz intraoperative ultrasound sector image. A finite recurrent sequence indexed by the first fourteen primes generates elliptic-integral signatures. The second-kind complete elliptic integral supplies an analytic energy corridor at every finite step. In the reference simulation (1.2 mm deformation amplitude, phase 0.4 rad), the model reports a target registration error (TRE) of 0.220 mm and Dice coefficient of 0.9887. These values are simulator outputs only; they do not establish clinical performance and require prospective phantom and patient-study validation.

1. Finite recurrent prime-moduli construction

Let p_n denote the n-th prime, with n = 1, ..., N and N = 14. Starting from k_0 = 0.15, the bounded modulus is updated at each finite step:

$$k_n = \left(k_{n-1} + \frac{1}{p_n}\right) \bmod 1, \quad 0 < k_n < 1, \quad p_n \in \{2, 3, 5, \ldots, 43\}$$

The complete elliptic integrals are evaluated with parameter m_n = k_n^2: K(k_n) = integral_0^(pi/2) [1 - k_n^2 sin^2(theta)]^(-1/2) dtheta and E(k_n) = integral_0^(pi/2) [1 - k_n^2 sin^2(theta)]^(1/2) dtheta.

2. Exact identity and finite energy bounds

For k'_n = sqrt(1 - k_n^2), the numerical computation is audited using Legendre's relation. The model uses the monotonic interval 1 <= E(k_n) <= pi/2 to normalize energy without extrapolating outside this analytic bound.

$$E(k_n)K(k'_n) + E(k'_n)K(k_n) - K(k_n)K(k'_n) = \frac{\pi}{2}$$

$$e_n = E_{\max}\frac{E(k_n)}{\pi/2}, \qquad \frac{2}{\pi}E_{\max} \le e_n \le E_{\max}$$

$$\frac{2}{\pi}nE_{\max} \le \sum_{i=1}^{n} e_i \le nE_{\max}, \qquad n \in \{1, \ldots, 14\}$$

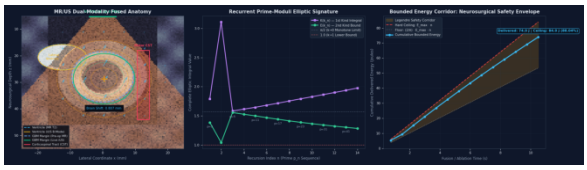
3. Target-registration characterization

The demonstrator reports a deterministic synthetic target-registration characteristic that rises with simulated displacement u: TRE = max(0.12, 0.18 + 0.05||u|| + 100 mean(|Delta_Legendre|)) mm. Segmentation agreement is reported as a simulated Dice score. These equations define application behavior rather than a calibrated biophysical error model.

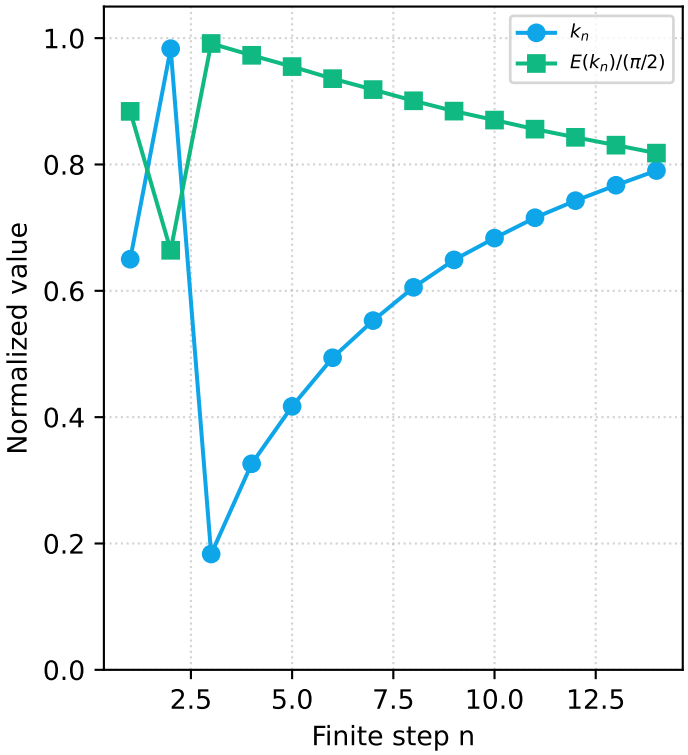
Disclosure: all images, anatomy, deformation, ultrasound speckle, and registration metrics in this preprint are computational simulations.

Synthetic Dual-Modality Fusion and Finite Signature Plots

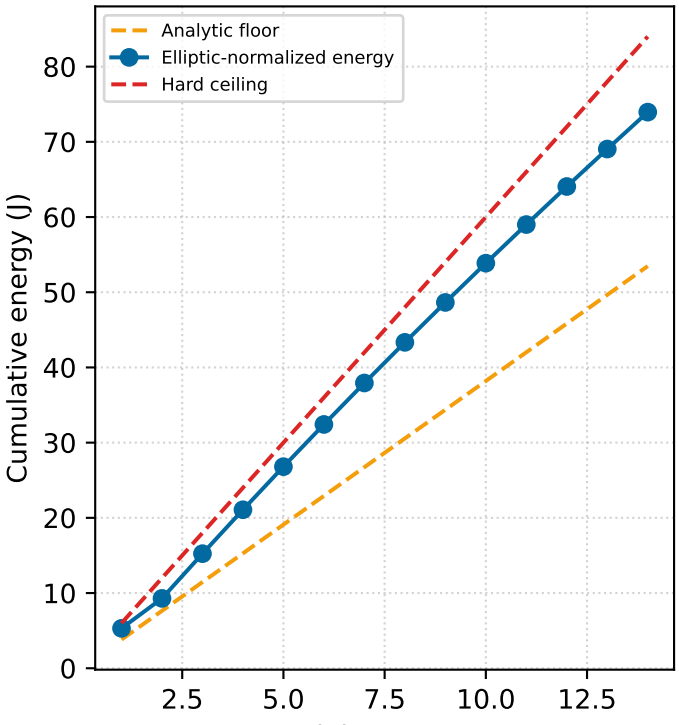
a | Fusion engine composite output



b | Prime-indexed modulus signature



c | Bounded cumulative energy corridor



d | Complete elliptic-integral characteristics

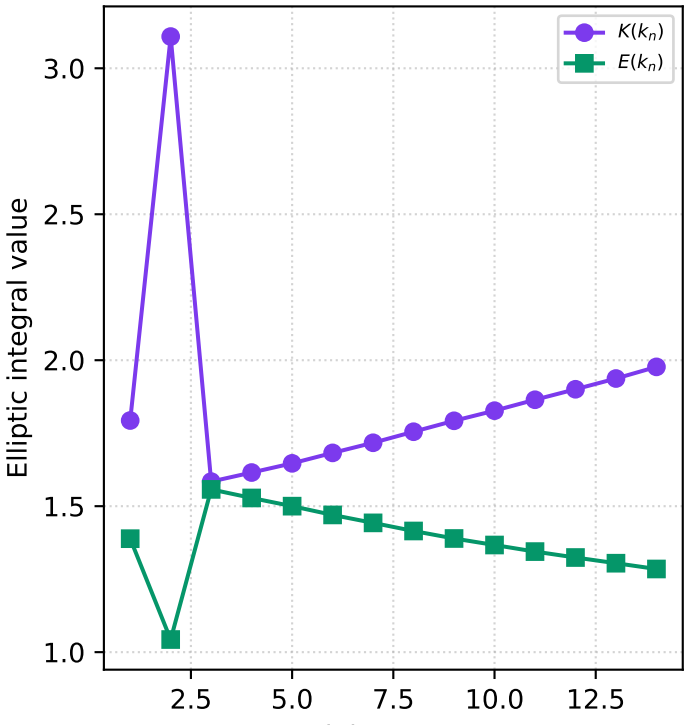


Figure 1 | Simulated MR/US anatomy is rendered with MRI and ultrasound overlays, deformable tissues, and a corticospinal-tract safety corridor. Panels b-d use the finite N = 14 prime sequence and E_max = 6 J reference setting.

Reference Simulation Characteristics and Finite Audit Schedule

Reference configuration

Characteristic	Reference value
MR modality	Synthetic pre-operative T2-weighted structural image
US modality	Synthetic 8.5 MHz sector B-mode image with Rayleigh speckle
Finite sequence	N = 14; seed k_0 = 0.15
Deformation input	1.2 mm amplitude; phase = 0.4 rad
Pulse model	E_max = 6.0 J; dt = 0.75 s
Synthetic TRE	0.220 mm
Synthetic Dice	0.9887
CST clearance	0.67 mm
Legendre max residual	8.882e-16
Cumulative energy	73.950 J / 84.000 J ceiling

Finite prime-moduli audit schedule

n	p_n	k_n	K(k_n)	E(k_n)	e_n (J)
1	2	0.6500	1.7935	1.3887	5.304
2	3	0.9833	3.1087	1.0432	3.985
3	5	0.1833	1.5843	1.5575	5.949
4	7	0.3262	1.6153	1.5281	5.837
5	11	0.4171	1.6467	1.5001	5.730
6	13	0.4940	1.6825	1.4701	5.615
7	17	0.5528	1.7172	1.4428	5.511
8	19	0.6055	1.7550	1.4150	5.405
9	23	0.6490	1.7925	1.3893	5.307
10	29	0.6834	1.8272	1.3670	5.222
11	31	0.7157	1.8646	1.3445	5.136
12	37	0.7427	1.9005	1.3242	5.058
13	41	0.7671	1.9373	1.3047	4.983
14	43	0.7904	1.9772	1.2849	4.908

Reproducibility note: values are deterministically computed from the source fusion engine using SciPy complete elliptic integrals. The finite energy bound is an analytic property of the normalization; it is not a clinical dose authorization or a substitute for safety validation.